

Joe Miguel Robertazzi

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EDUCATION

Stanford University

Candidate for Bachelor of Science in Computer Science

Palo Alto, CA

Sep. 2023 – June 2027

GPA: 3.9 / 4.0

Relevant Coursework: Circuits, Computer Systems from the Ground Up, Calculus, Data Structures, Discrete Mathematics, Probability, Proof Writing, Linear Algebra & Multivariable Calculus, Diff. Eq. with Linear Algebra & Fourier Methods

SKILLS

Programming: C/C++, Java, Python, JavaScript | **Libraries:** NumPy, Pandas, scikit-learn, Matplotlib, Seaborn, PyTorch

WORK EXPERIENCE

Stanford University

Teaching Assistant, Department of Computer Science

Palo Alto, CA

Apr. 2024 – Present

- Conduct weekly 10+ person discussion sections of Stanford's introductory computer science courses CS106A & CS106B
- Oversee biweekly grading sessions with students to review coding errors from assignments, ensuring mastery over material
- Debug code with students 1-on-1 during office hours and graded midterm and final exams alongside other section leaders

University of Rochester

Independent Student Researcher

Rochester, NY

Jan. 2021 – June 2023

- Developed a [research project](#) investigating the relationship between Earth's magnetic field and migratory bird patterns
- Conducted innovative analysis of datasets with 50+ million entries through novel map-based data visualizations via Python
- Wrote a professional research manuscript and presented findings; won at national competitions, including Regeneron STS

PROJECTS

Leonardo DaVinci – Automated LEGO Image Printer | Python, C, Assembly

Jan. 2024 – Mar. 2024

- Designed, built, and programmed a bare-metal device that prints user-loaded images via LEGO blocks.
- Developed a complete back-end system in C and Python, leveraging a RISC-V MangoPi for processing.
- Rewired and integrated sensors/motors to precisely position LEGO pieces in 4.8mm x 4.8mm slots

Shell Terminal – | Assembly, C

Jan. 2024 – Feb. 2024

- Designed and implemented shell terminal with custom commands on RISC-V MangoPi system.
- Integrated keyboard input using interrupt handling and a graphical interface via custom-developed graphics library

Research Project Analyzing Avian Migratory Patterns | Python, scikit-learn, Pandas, Seaborn

Jan. 2020 – June 2023

- Developed 100+ interactive geospatial maps using libraries like pygmt and Seaborn to visualize migration data.
- Designed and implemented 4 analytical frameworks through Python and scikit-learn to elucidate correlations between bird migration, magnetic fields, and environmental factors.
- Identified strong correlations ($r > 0.7$) between declining bird populations and weakening magnetic fields.

Ossining Engineering Club Summer Program | PowerPoint, C++, FRC Robotics

June 2022 – June 2023

- Established an 8-week long curriculum focused on C++ fundamentals and object-oriented programming (OOP)
- Empowered financially disadvantaged individuals by holding free weekly classes virtually via Google Meets and Zoom
- Successfully attracted 10+ students per week, with 100% completing a robot design project in CAD/C++

LEADERSHIP

Ossining Engineering Club / FRC Team 4122

Vice President & Treasurer

Ossining, NY

Sep. 2019 – June 2023

- Led programming efforts, earning the team two coding awards and securing a spot in the FRC World Championship
- Coordinated build-season meetings and mentored 45+ members across mechanical and programming departments
- Developed a real-time statistical framework using NumPy to analyze competitor performance and refine playoff strategies.

ADDITIONAL INFORMATION

Interests: Running, Hip-Hop/J-Pop, Dystopian Novels, Robotics, Computer Building, Competitive Online Games

Soft skills: Adaptable, Analytical thinking, Collaboration, Communication, Decisive, Open-minded, Problem solving